



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2017

CAL LAB LIMITED
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CALIBRATION

Valid To: September 30, 2023

Certificate Number: 3815.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations^{1, 7}:

I. Chemical

Parameter/Equipment	Range ⁵	CMC ^{2, 4} (±)	Comments
Gas Detection Equipment –			
Carbon Monoxide	(0 to 500) ppm	2.2 % FS	Reference gas comparison BS EN 60079-29-2/ JJG 635/ JJG915
Carbon Dioxide	(0 to 5000) ppm	2.0 % FS	Reference gas comparison BS EN 60079-29-2/ JJG 635
Isobutylene	(0 to 10) ppm	3.6 % FS	Reference gas comparison BS EN 60079-29-2/ JJF 1172
Hydrogen Sulfide	(0 to 100) ppm	6.6 % FS	Reference gas comparison BS EN 60079-29-2/ JJG 695
Methane	(0 to 5) %	2.6 % FS	Reference gas comparison BS EN 60079-29-2/ JJG 693

Parameter/Equipment	Range ⁵	CMC ^{2, 4, 6} (±)	Comments
Gas Detection Equipment – (cont)			
Oxygen	6 % Volume 15 % Volume 24 % Volume	0.45 % 0.98 % 1.5 %	Reference gas comparison BS EN 60079-29-2/ JJG 365
Nitrogen Dioxide	(0 to 1) ppm	2.4 % FS	Reference gas comparison BS EN 60079-29-2
Ozone	(0.05 to 1) ppm (1 to 100) ppm	2.8 % FS 3.4 % FS	Reference gas comparison BS EN 60079-29-2/ JJG 1077
Sulfur Dioxide	(0 to 20) ppm	7.2 % FS	Reference gas comparison BS EN 60079-29-2/ JJG 551
Formaldehyde	(0 to 1) ppm	6.2 % FS	Reference gas comparison BS EN 60079-29-2/ JJG 1022
Radon	(500 to 4000) Bq/ m ³	17 % FS	Reference meter comparison JJG 825
Dust Concentration			
TSP	(0 to 10) mg/m ³	14 % FS	Reference meter comparison ISO 21501-4/ JJF 1659/ JJG 846
PM10	(0 to 10) mg/m ³	17 % FS	
PM2.5	(0 to 10) mg/m ³	17 % FS	

II. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2, 3, 6} (±)	Comments
DC Current – Measure	(0.1 to 2) mA (2 to 20) mA (20 to 200) mA (0.2 to 0.5) A (0.5 to 1) A	0.080 % 0.080 % 0.080 % 0.078 % 0.18 %	Keithley 2100 digital multimeter

Parameter/Equipment	Range	CMC ^{2, 3, 6} ($\pm$)	Comments
DC Voltage – Measure	(0.25 to 1) V (1 to 10) V (10 to 100) V (100 to 500) V (500 to 1000) V	0.0072 % 0.0064 % 0.0082 % 0.010 % 0.0098 %	Keithley 2100 digital multimeter
DC Resistance – Measure	(0.1 to 100) Ω (0.1 to 1) k Ω (1 to 10) k Ω (10 to 100) k Ω (0.1 to 1) M Ω (1 to 10) M Ω	0.026 % 0.023 % 0.020 % 0.023 % 0.025 % 0.055 %	Keithley 2100 digital multimeter
DC Current – Generate	(0.1 to 2) mA (2 to 20) mA (20 to 200) mA (0.2 to 0.5) A (0.5 to 1) A	0.15 % 0.14 % 0.15 % 0.12 % 0.20 %	Huaguang DO30-E+ digital calibrator & Keithley digital multimeter
DC Voltage – Generate	(0.25 to 1) V (1 to 10) V (10 to 100) V (100 to 500) V (500 to 1000) V	0.088 % 0.11 % 0.11 % 0.11 % 0.11 %	Huaguang DO30-E+ digital calibrator & Keithley digital multimeter
DC Resistance – Generate	10 Ω 100 Ω 1 k Ω 10 k Ω 100 k Ω 1 M Ω 10 M Ω	0.25 % 0.26 % 0.24 % 0.23 % 0.23 % 0.24 % 0.24 %	Huaguang DO30-E+ digital calibrator & Keithley digital multimeter

Parameter/Range	Frequency	CMC ^{2, 3, 6} (±)	Comments
AC Current – Measure (0.1 to 2) mA (2 to 20) mA (20 to 200) mA (0.2 to 0.5) A (0.5 to 1) A	60 Hz	0.80 % 0.32 % 0.30 % 0.27 % 0.27 %	Keithley 2100 digital multimeter
AC Voltage – Measure (0.25 to 1) V (1 to 10) V (10 to 100) V (100 to 500) V (500 to 1000) V	60 Hz	0.19 % 0.20 % 0.20 % 0.21 % 0.19 %	Keithley 2100 digital multimeter
AC Current – Generate (2 to 20) mA (20 to 200) mA (0.2 to 0.5) A (0.5 to 1) A	60 Hz	0.65 % 0.37 % 0.32 % 0.31 %	Huaguang DO30-E+ digital calibrator and Keithley digital multimeter
AC Voltage – Generate (0.25 to 1) V (1 to 10) V (10 to 100) V (100 to 500) V (500 to 1000) V	60 Hz	0.32 % 0.28 % 0.28 % 0.30 % 0.28 %	Huaguang DO30-E+ digital calibrator and Keithley digital multimeter

III. Fluid Quantities

Parameter/Equipment	Range	CMC ^{2, 3, 6} (±)	Comments
Air Flow Rate – Measure	(0 to 30) L/min	3.0 %	Reference air flow comparison JJG 956 (Sec. 6.3.3.1)
Air Velocity – Measuring Equipment	(0 to 10) m/s	3.6 %	Reference anemometer

IV. Thermodynamics

Parameter/Equipment	Range	CMC ^{2,3} ($\pm$)	Comments
Temperature – Measuring Equipment	(-20 to 5) °C (5 to 55) °C (55 to 80) °C	0.58 °C 0.14 °C 0.58 °C	Reference thermometer JJF 1183
Liquid-In-Glass Thermometer	(-20 to 300) °C	0.14 °C	Reference thermometer JJG 130
Dial Thermometer	(-20 to 300) °C	1.2 °C	
Thermometer With Display Unit	(-20 to 300) °C	0.038 °C	
Humidity Detecting Equipment	(10 to 90) % RH	1.9 % RH	Reference humidity probe JJF 1076

¹ Field This laboratory offers commercial calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ The type of instrument or material being calibrated is defined by the parameter. This indicates the laboratory is capable of calibrating instruments that measure or generate the values in the ranges indicated for the listed measurement parameter.

⁴ In the statement of CMC, FS represents "Full Scale".

⁵ When "ppm" is used on the scope, it refers to parts in 10^6 .

⁶ In the statement of CMC, percentages are percentages of reading, unless otherwise indicated.

⁷ This scope meets A2LA's *P112 Flexible Scope Policy*.



Accredited Laboratory

A2LA has accredited

CAL LAB LIMITED

Tsuen Wan, NT, HONG KONG

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2017 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets R205 – Specific Requirements: Calibration Laboratory Accreditation Program. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated April 2017).



Presented this 1st day of November 2021.

A blue ink signature of the Vice President of Accreditation Services.

Vice President, Accreditation Services
For the Accreditation Council
Certificate Number 3815.01
Valid to September 30, 2023

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.